PROBLEM SET 6

Show **ALL WORK** to get full/partial credit. Begin each problem on a new page, and clearly label each part of the problem. (**Max Score: 45**)

1) **(10 pts)** Verity that $H_3(y)$ and $H_4(y)$ obey the recursion relation, Eq. (7.3.15)

2) **(20 pts)** Consider a particle in a potential for the general quantum oscillator problem which we solved in class, staring from Eq. 7.3.8, where $y = x\sqrt{m\omega/\hbar}$

(**general hint**: it is instructive to follow once again the general procedure from the text summarized in steps 1-4 on p. 196, and apply it to this potential)

$$\begin{aligned} V(x) &= \frac{1}{2}m\omega^2 x^2, \; x > 0 \\ &= \infty, \qquad\quad x \le 0 \end{aligned}$$

(a) **(5 pts)** Make a plot of the potential $V(x)$ vs. x

(b) **(5 pts)** What is the boundary condition on the wave function $\psi(x)$ as $x \to 0$?

(c) **(5 pts)** Using the result from part (b), show that only the n = odd solutions of $\psi_n(y) = u(y)e^{-y^2/2}$ are fulfilled (as $y \to 0$, truncate the terms of order $\mathcal{O}(y^2)$ and check which coefficients C_n in Eq. 7.3.19 survive based on the boundary condition)

(d) **(5 pts)** Show that the eigenvalues reduce to the form (remember the general expression $n = 2k + m$ on p. 194, where)

$$E_k = \left(2k + \frac{3}{2}\right)\hbar\omega = \frac{4k+3}{2}\hbar\omega, \;\; k = 0,1,2, \ldots$$

3) **(15 pts)** *The Oscillator in Momentum Space*. By setting up an eigenvalue equation for the oscillator in the P basis and comparing it to Eq. (7.3.2), show that the momentum space eigenfunctions may be obtained from the ones in coordinate space through the substitution $x \to p, \quad m\omega \to 1/m\omega$. Thus, for example, once you obtain the general solution $\psi_n(p)$, show that

$$\psi_0(p) = (1/m\pi\hbar\omega)^{1/4} \cdot e^{-p^2/2m\hbar\omega}$$

There are several other pairs, such as ΔX and ΔP in the state $|n\rangle$, which are related by the substitution $m\omega \to 1/m\omega$. You may wish to watch out for them. (Refer back to Exercise 7.3.5)